

Building an Intelligent Supply Chain Disruption Management Platform

Overview

Your team at **GlobalLogic** has been hired by **SecureTech Solutions**, a multinational cybersecurity company, to build an AI-powered Security Operations Center (SOC) Assistant.

Every day, security analysts receive hundreds of alerts from firewalls, endpoint protection systems, cloud infrastructure, authentication services, and employee reports.

Analysts spend significant time reviewing alerts, identifying false positives, gathering related information, escalating incidents, and preparing investigation reports.

SecureTech wants a single AI assistant capable of understanding security incidents, collecting relevant information, routing requests to the appropriate AI workflows, and assisting analysts throughout the investigation process.

The client expects a production-ready MVP within **two days**.

You will work as an AI consulting team, where each member is responsible for a different part of the solution.

Client Profile

Organization

SecureTech Solutions

- Global Security Operations Center
- 5 Regional SOC Teams
- 20,000 Protected Endpoints
- Cloud Infrastructure
- Identity Management Platform
- Enterprise Monitoring Systems

Existing Technology

- SIEM Platform
- Endpoint Protection System
- Identity & Access Management
- Firewall Monitoring
- Threat Intelligence Portal
- Incident Management System
- Email & Teams

All systems expose mock REST APIs or JSON data for this capstone. It will be provided with the project.

Current Business Challenges

Security analysts currently use multiple systems.

Examples:

- Search Security Alerts
- Check Login Activity
- Review Endpoint Status
- Create Security Incident
- Investigate Suspicious IP
- Review User Activity
- Generate Incident Reports

Security teams want one AI assistant instead of switching between multiple dashboards.

Executive Goals

The CISO expects:

- Faster Incident Investigation
- Reduced Alert Fatigue
- Intelligent Alert Routing
- Improved Analyst Productivity

- AI-Assisted Security Operations
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Project Objective

Build an AI-powered Security Operations Assistant capable of understanding security requests, deciding which security function should handle them, executing the required workflow, and responding naturally.

Team Structure and Role

Students should work in teams of **4 members**.

Team Member 1

AI Conversation Engineer

Responsible for:

- Chat UI
 - Prompt Engineering
 - Memory
 - Conversation History
 - System Prompt
 - User Experience
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Team Member 2

Tool & Integration Engineer

Responsible for building tools such as:

- Alert Lookup
- User Activity Search
- Device Status Checker

- Login History
 - Threat Intelligence Lookup
 - Incident Creator
 - Report Generator
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Team Member 3

Agent Engineer

Responsible for:

- LangGraph Workflow
 - Agent Logic
 - Tool Selection
 - Human-in-the-loop
 - Error Handling
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Team Member 4

Multi-Agent Engineer

Responsible for:

- Supervisor Agent
 - Alert Analysis Agent
 - Endpoint Agent
 - Identity Agent
 - Incident Agent
 - Reporting Agent
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All Members

Evaluation & Deployment Engineer

Responsible for:

- LangSmith Tracing

- Prompt Evaluation
 - Testing
 - Deployment
 - Documentation
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Functional Requirements

The assistant should support requests like:

Security Alerts

- Search Alerts
- Check Alert Severity
- Review Alert Details

Identity

- Check Login History
- Review Failed Login Attempts
- Search User Activity

Endpoint Security

- Check Device Status
- Review Malware Detection
- Verify Device Health

Incident Management

- Create Security Incident
- Check Incident Status
- Escalate Incident

Reporting

- Generate Investigation Report
- Summarize Security Alerts
- Prepare Executive Summary

Level 1

AI Assistant

Build a conversational Security Operations assistant.

Features

- Chat UI
 - Memory
 - Prompt Engineering
 - Guardrails
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Level 2

AI Agent

Convert the chatbot into an AI Agent capable of invoking tools.

Example tools

`search_alert()`

`search_user()`

`check_endpoint()`

`lookup_ip()`

`create_incident()`

`generate_report()`

☐ The agent should automatically determine which tool(s) to invoke.

Level 3

Multi-Agent Architecture

Build a LangGraph-based workflow.

Recommended Agents

Supervisor Agent

Receives every request.

Determines:

- Alerts
- Identity
- Endpoint
- Incident
- Reporting

Routes the request accordingly.

Alert Analysis Agent

Handles:

- Security Alerts
 - Alert Severity
 - Threat Summary
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Endpoint Agent

Handles:

- Device Health
- Malware Detection
- Endpoint Information

Identity Agent

Handles:

- Login History
- User Activity
- Authentication Events

Incident Agent

Handles:

- Incident Creation
- Incident Investigation
- Escalation

Reporting Agent

Handles:

- Investigation Report
- Executive Summary
- Incident Timeline

Human-in-the-loop

The system must request confirmation before:

- Creating an Incident
 - Escalating an Incident
 - Marking an Alert as Critical
 - Closing an Investigation
 - Generating a Final Report
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LangSmith Requirements

Every team must:

- Enable tracing
 - Record at least 10 conversations
 - Analyze latency
 - Review failed runs
 - Improve one prompt based on trace insights
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Deployment

Deploy the application using:

- Streamlit
 - Environment Variables
 - Production-ready README
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Deliverables

Each team must submit:

1. Working Application

Fully functional Streamlit application.

2. Source Code

Well-structured repository with proper folder organization.

3. Architecture Diagram

Illustrate:

- LLM

- Tools
 - LangGraph Workflow
 - Multi-Agent Design
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4. LangSmith Report

Include:

- Traces
 - Prompt Improvements
 - Evaluation Observations
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5. Deployment

Live URL (using a free cloud service like Streamlit) or a local deployment demonstration if internet access is unavailable.

6. Technical Presentation

Include:

- Project Overview
 - Setup Instructions
 - Folder Structure
 - Team Responsibilities
 - Future Enhancements
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PHASE 2 CLIENT ENHANCEMENT

SecureOps AI

Version 2 – Adaptive SOC Copilot with Threat Intelligence

Background

After the successful deployment of SecureOps AI, analysts reported faster incident investigations. However, attackers continuously evolve their techniques, tactics and infrastructure. Indicators of compromise (IOCs), attacker behaviors and vulnerability information change daily, making static investigation playbooks insufficient.

SecureTech Solutions now wants SecureOps AI to evolve into an Adaptive SOC Copilot that continuously incorporates threat intelligence and assists analysts with up-to-date investigation guidance.

Business Context

The Security Operations Center receives threat intelligence from commercial feeds, open-source intelligence, vulnerability advisories and internal investigations. Analysts struggle to correlate this external intelligence with ongoing incidents, resulting in missed attack indicators and delayed responses. The organization requires an AI assistant that can reason over current intelligence and adapt investigation recommendations dynamically.

Phase 2 Client Requirement

Enhance SecureOps AI to consume threat intelligence, evaluate its relevance to ongoing investigations, enrich alerts with contextual information and recommend the next investigative steps. The assistant should explain why a threat is relevant, identify affected assets and recommend defensive actions while allowing security analysts to approve or reject recommendations.

Illustrative Security Scenarios

- New ransomware campaign targets software used inside the organization.
- A zero-day vulnerability affects externally exposed servers.
- Threat intelligence identifies a malicious domain communicating with employee devices.

- A phishing campaign targets executives using newly registered look-alike domains.
- An internal alert matches tactics described in a recent MITRE ATT&CK report.
- Previously low-risk alerts become high priority because of newly published intelligence.
- Multiple incidents share infrastructure linked to the same threat actor.

Business Objectives

- Continuously enrich investigations with current threat intelligence.
- Prioritize incidents based on emerging risks.
- Improve analyst decision-making using explainable AI.
- Reduce investigation time through contextual recommendations.
- Support rapid adaptation to evolving attacker techniques.

Architecture Exploration

Teams should explore architectures that combine enterprise telemetry with external threat intelligence. Consider intelligence ingestion, knowledge enrichment, confidence scoring, contextual reasoning, analyst feedback loops and continuous learning. The implementation is intentionally open; students should justify how their design remains scalable, explainable and secure.

Design Challenges

- How should trusted and untrusted intelligence sources be handled?
- How should conflicting intelligence reports be resolved?
- How can confidence scores influence recommendations?
- How should stale threat intelligence be retired?
- How should analysts provide feedback to improve future recommendations?

Expected Deliverables

1. Extended architecture illustrating threat intelligence integration.
2. Demonstration of five investigations enriched using external intelligence.
3. Design justification covering enrichment, prioritization and confidence scoring.
4. Discussion of governance, scalability and future enhancements.

Evaluation Expectations

Assessment will emphasize architectural reasoning, effective use of threat intelligence, explainability, analyst-centric design and the ability to justify security trade-offs. Multiple architectural approaches are acceptable provided they address the enterprise objectives.

Evaluation Criteria

Criteria	Weight(100)
Prompt Engineering & Conversation Design	15
LangChain & Tool Integration	20
Agent & LangGraph Workflow	20
Multi-Agent Collaboration	15
LangSmith Tracing & Evaluation	10
Deployment & Documentation	10
Team Collaboration & Final Presentation	10